

Functional Annotation of *acnA-II* (PP_2336, UniProt Q88KF4) in *Pseudomonas putida* KT2440

Target: Gene *acnA-II*; OrderedLocusName **PP_2336**; UniProt **Q88KF4** (TrEMBL, unreviewed; protein existence = inferred from homology). **Organism:** *Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / NCIMB 11950 / **KT2440**). **Report date:** 2026-07-23.

1. Summary (answer to the research question)

PP_2336 encodes a **cytoplasmic, [4Fe-4S]-cluster-dependent dehydratase of the aconitase superfamily**. Although the automated gene symbol ("*acnA-II*") and the generic UniProt label ("aconitate hydratase, EC 4.2.1.3") suggest an ordinary tricarboxylic-acid (TCA)-cycle aconitase, the ortholog assignment, protein-family signature, and — most decisively — its position within the *P. putida* **methylcitrate (prp) operon** identify it as an **AcnD-type 2-methylcitrate dehydratase (EC 4.2.1.117; KEGG K20455)**. Its specific, committed role is the dehydration of **(2S,3S)-2-methylcitrate to 2-methyl-*trans*-aconitate** (and, by the same aconitase-like half-reaction, citrate → *cis*-aconitate) in the **2-methylcitrate cycle** that catabolises propionate (and propionyl-CoA from odd-chain fatty acids / amino-acid degradation) to pyruvate and succinate. It works obligately with the adjacent accessory isomerase **PrpF (PP_2337)**, which converts the *trans* product to 2-methyl-*cis*-aconitate for downstream hydration by the housekeeping aconitase **AcnB (PP_2339)**.

2. Molecular identity and biophysical properties

Property	Value	Source
Length / mass	862 aa / ~93.96 kDa, monomeric	UniProt Q88KF4
Cofactor	one [4Fe-4S] cluster	UniProt Q88KF4 (keywords: 4Fe-4S, Iron, Iron-sulfur, Metal-binding)
Domains	Aconitase/3-IPM-dehydratase large $\alpha/\beta/\alpha$ (res 65–533); Aconitase A "swivel" small-subunit domain (res 656–788)	UniProt features; Pfam PF00330 + PF00694
Family signatures	IPR012708 "2-methylisocitrate dehydratase AcnD, Fe/S-dependent" ; IPR006249 "Aconitase/IRP2"; IPR001030/IPR015928/IPR015931/IPR000573; COG1048	InterPro / eggNOG
Orthology	KEGG K20455 — 2-methylcitrate dehydratase (2-methyl-<i>trans</i>-aconitate forming), EC 4.2.1.117	KEGG ppu:PP_2336
Localization	Cytoplasmic (no signal peptide/TM segment; universal for aconitase-family enzymes)	Inferred (sequence + family)
Evidence level	Inferred from homology (PE3); no structure or direct assay for PP_2336 itself	UniProt

All members of the aconitase superfamily are soluble **cytoplasmic** enzymes that carry a catalytic [4Fe-4S] cluster in which three iron atoms are Cys-ligated and the fourth (Fe-a) is solvent-exposed and coordinates the substrate hydroxyl/carboxylate during dehydration-rehydration [Lauble 1994,

P 8151704

Bacterial aconitases are "distantly related **monomeric Fe-S proteins** that contain different arrangements of four structural domains" [Jordan 1999,

P 10585860

— exactly the two-lobe (large $\alpha/\beta/\alpha$ + swivel) architecture annotated for PP_2336.

3. Primary function: the reaction catalysed and substrate specificity

PP_2336 catalyses the dehydration of 2-methylcitrate to 2-methyl-*trans*-aconitate (EC 4.2.1.117), and the analogous aconitase-like dehydration of citrate to *cis*-aconitate / hydration of *cis*-aconitate. It does **not** carry out the second (isocitrate-forming) half of the classical aconitase reaction.

The definitive biochemistry comes from the experimentally characterised AcnD orthologs (*Shewanella oneidensis* MR-1 and *Vibrio cholerae*), whose sequences, KO (K20455) and family signature (IPR012708/COG1048) are shared by PP_2336:

- "AcnD dehydrated 2-methylcitrate and citrate to 2-methyl-*cis*-aconitate and *cis*-aconitate, respectively; AcnD also hydrated *cis*-aconitate. However, 2-methylisocitrate and isocitrate were not substrates for AcnD, indicating that AcnD only catalyzes the first half of the aconitase-like dehydration reactions." [Grimek & Escalante-Semerena 2004,

P 14702315]

- "The AcnD protein required reconstitution of an Fe/S cluster for activity. All detectable AcnD activity was lost after incubation with iron-chelating agents..." [PMID 14702315] — consistent with the [4Fe-4S] cofactor annotated for Q88KF4.
- The KEGG EC 4.2.1.117 designation ("2-methyl-*trans*-aconitate forming") and later mechanistic work show the AcnD product is the *trans* (not *cis*) isomer, which is why a dedicated isomerase (PrpF) is required downstream [Rocco et al. 2017,

P 29145506].

Substrate specificity, precisely stated: tricarboxylate substrates bearing a C2 hydroxyl and (for the physiological substrate) a 2-methyl branch — i.e. **(2S,3S)-2-methylcitrate** (physiological) and **citrate** (promiscuous) — undergoing *anti/syn* elimination of water to the corresponding aconitate. The enzyme is specialised for the **dehydration (first) half-reaction only**.

4. Pathway context: the AcnD–PrpF variant of the 2-methylcitrate cycle

PP_2336 lies in a contiguous, co-oriented **prp/methylcitrate gene cluster** in KT2440 (KEGG module **M00982 Methylcitrate cycle**; pathway **ppu00640 Propanoate metabolism**; GO:0019679 "propionate metabolic process, methylcitrate cycle"):

Locus	Gene / enzyme	KO / EC
PP_2333	GntR-family regulator (PrpR-like)	—
PP_2334	2-methylisocitrate lyase (PrpB)	K03417 / 4.1.3.30
PP_2335	2-methylcitrate synthase (PrpC)	K01659 / 2.3.3.5
PP_2336	AcnD — 2-methylcitrate dehydratase (this gene)	K20455 / 4.2.1.117
PP_2337	2-methyлаconitate isomerase (PrpF)	K09788 / 5.3.3.–
PP_2338	2-methylcitrate dehydratase (PrpD)	K01720 / 4.2.1.79
PP_2339	bifunctional aconitate hydratase 2 / 2-methylisocitrate dehydratase (AcnB)	K01682 / 4.2.1.3, 4.2.1.99

Reaction sequence of the cycle (propionate entry): 1. Propionyl-CoA + oxaloacetate → **2-methylcitrate** (PrpC, PP_2335). 2. **2-methylcitrate** → **2-methyl-*trans*-aconitate** — **AcnD (PP_2336)**. 3. 2-methyl-*trans*-aconitate → 2-methyl- *cis*-aconitate — **PrpF (PP_2337)**. 4. 2-methyl-*cis*-aconitate → **(2R,3S)-2-methylisocitrate** — hydration by **AcnB (PP_2339)**. 5. 2-methylisocitrate → **pyruvate + succinate** — PrpB (PP_2334).

The obligatory hand-off is established directly: "**We show that the SoPrpF protein isomerizes the product of the AcnD reaction into the PrpD product (2-MCA), a known substrate of the housekeeping aconitase (AcnB)**" [Rocco et al. 2017, [P 29145506](#)],

and the AcnD+PrpF pair is functionally sufficient: "**the AcnD and PrpF proteins restored the ability of a prpD mutant strain of *Salmonella enterica* to grow on propionate as a source of carbon and energy**" [Grimek & Escalante-Semerena 2004, [P 14702315](#)].

Why two routes? KT2440 encodes *both* an Fe/S-independent PrpD (PP_2338; forms 2-methyl-*cis*-aconitate directly [Horswill & Escalante-Semerena 2001,

P 11294638

Brock et al. 2002,

P 12473114

and the AcnD–PrpF pair. These provide **parallel/redundant 2-methylcitrate-dehydration capacity**; the AcnD route (Fe/S-dependent) is thought to predominate or be favoured under particular redox/iron conditions, whereas PrpD (Fe/S-independent) is robust to oxidative stress. This redundancy explains why *acnA/acnD* single mutants in related organisms often show no propionate-growth defect [Horswill 2001,

P 11294638

Biological significance: the methylcitrate cycle is the principal aerobic route for catabolising **propionate/propionyl-CoA** — derived from odd-chain fatty acids, branched-chain amino acids (Ile, Val, Met, Thr), cholesterol side chains and environmental short-chain acids — and for **detoxifying** the growth-inhibitory intermediate propionyl-CoA in this soil bacterium. This is directly relevant to KT2440's role as a metabolically versatile bioremediation/biotechnology chassis.

5. Distinction from the housekeeping TCA aconitases

P. putida KT2440 encodes several aconitase-superfamily proteins with distinct jobs (KEGG EC linkage): - **PP_2112** and **PP_2339 (AcnB)** — EC 4.2.1.3 / 4.2.1.99, the **canonical TCA-cycle aconitases** (full citrate $\rightleftharpoons$ isocitrate isomerisation) and AcnB additionally the 2-methylisocitrate dehydratase. - **PP_2338 (PrpD)** — EC 4.2.1.79, Fe/S-independent 2-methylcitrate dehydratase. - **PP_2336 (this gene)** — **the unique EC 4.2.1.117 / AcnD**, specialised for the *trans*-forming first half-reaction in the methylcitrate cycle.

Thus PP_2336 is **not** the enzyme that supplies isocitrate to the TCA cycle; that role belongs to PP_2112/PP_2339. The bacterial paradigm in which two aconitases (AcnA = stationary-phase/stress-induced, more stable, moonlighting mRNA-binding; AcnB = major exponential-phase enzyme) partition function and can act as post-transcriptional regulators [Tang & Guest 1999,

P 10589714

Tang et al. 2002,

[P 11932448](#)

Baothman et al. 2013,

[P 23637460](#)

is informative background, but PP_2336 belongs to the **methylcitrate-dedicated AcnD subfamily**, a functionally separate branch. There is no evidence that PP_2336 itself moonlights as an RNA-binding regulator.

6. Localization

The enzyme functions in the **cytoplasm**. Aconitase-superfamily proteins are soluble; the sequence lacks a signal peptide or transmembrane region, and the methylcitrate-cycle intermediates it processes are cytosolic. No experimental localization data exist for PP_2336 specifically; this assignment is by strong homology and biochemical logic.

7. Supported and refuted hypotheses

Supported - H1 — PP_2336 is an aconitase-superfamily [4Fe-4S] dehydratase. Supported by UniProt cofactor annotation, InterPro/Pfam signatures, COG1048. - **H2 — Its specific/primary function is AcnD-type 2-methylcitrate dehydratase (EC 4.2.1.117), forming 2-methyl-*trans*-aconitate.** Strongly supported by KEGG K20455, InterPro IPR012708, and the *prp*-operon context; mechanism validated in orthologs [PMID 14702315; 29145506]. - **H3 — It operates in the 2-methylcitrate cycle for propionate catabolism, obligately partnered with PrpF (PP_2337) and feeding AcnB (PP_2339).** Supported by operon structure, KEGG module M00982, and ortholog reconstitution/complementation studies.

Refuted / not supported - H4 — PP_2336 is a canonical TCA-cycle aconitase interconverting citrate and isocitrate. Refuted: AcnD enzymes do not use isocitrate/2-methylisocitrate as substrates (only the first half-reaction) [PMID 14702315]; the TCA aconitase role in KT2440 is filled by PP_2112/PP_2339. The "acnA-II"/"EC 4.2.1.3" labels are automated over-generalisations. - **H5 — PP_2336 is a moonlighting mRNA-binding regulator (IRP/AcnA-type).** Not supported for this subfamily; no evidence.

8. Limitations and future directions

- **Evidence is inferential for PP_2336 itself** (UniProt PE3 "inferred from homology"; unreviewed TrEMBL). All enzymology derives from AcnD/PrpF orthologs in *Shewanella*, *Vibrio*, and *Salmonella*, plus the KT2440 genomic context — no purified-enzyme assay, kinetics, or structure exists for Q88KF4.
- **Recommended validation:** (i) heterologous expression + in-vitro assay of PP_2336 with 2-methylcitrate/citrate (product stereochemistry by NMR) and Fe/S reconstitution; (ii) Δ PP_2336, Δ PP_2337, Δ PP_2338 and combinatorial mutants tested for growth on propionate vs glucose to define redundancy with PrpD; (iii) transcriptional analysis of the *prp* operon (PP_2333 PrpR/2-methylcitrate induction); (iv) an AlphaFold model + docking to confirm the [4Fe-4S] active site and *trans*-forming geometry.
- **Open mechanistic question:** the precise identity of the AcnD product isomer (2-methyl-*trans*-aconitate vs 4-methylaconitate) and the exact accessory role of PrpF (isomerase vs Fe/S-protecting) remain debated even in the model organisms [Rocco et al. 2017,

P 29145506

9. Key references

- Grimek TL, Escalante-Semerena JC. *J Bacteriol* (2004). AcnD is a new Fe/S-dependent 2-methylcitrate dehydratase requiring PrpF.

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P 42464328

- Database records: UniProt **Q88KF4**; KEGG **ppu:PP_2336** (K20455); InterPro **IPR012708/ IPR006249**; Pfam PF00330/PF00694; eggNOG **COG1048**.